

Modulation Test Tool

HF Channel Sounder for the Hermes-Lite 2 — how each waveform is transmitted and received
Same tool both ends of the link • RF 7.13 MHz • fs 48 ksps • 5 s grabs

The workflow, and the one rule behind it

One radio transmits, the other receives. On the TX radio, pick a Signal type and key up (the master TX bar keys Tone + PTT + En PA together). On the RX radio, pick the **same** Signal type and press **Grab + Analyze**; it captures a fixed window (GRAB_SEC, 5 s by default) and analyses it.

The TX side loops its waveform continuously the entire time it is keyed — back to back, no gaps. **The RX side grabs a fixed window at any moment** and the analyser locates the waveform inside that window by itself. That is why the two operators never have to synchronize start times: TX keys up, then the other operator presses Grab whenever. Looping is what buys that freedom.

Two requirements: both radios must be on the **same mode** (the receiver needs the matching reference to demodulate), and the TX must be keyed before/during the grab. Longer grabs average more repetitions and resolve Doppler more finely. Captured files are written to the **capture** folder as `<mode>_analysis.txt` and `<mode>_data.npz`.

Three families of waveform

The five modes recover the channel in fundamentally different ways:

- **Correlation / matched-filter** (LFM chirp, DSSS/PN) — a short pulse or code is repeated continuously; the RX correlates every repetition against a known reference, which lifts a sub-noise signal by the processing gain and builds the channel impulse response. Inherently tolerant of a carrier-frequency offset (it only slides the correlation peak).
- **Frequency-domain probes** (Comb, OFDM) — a known spectrum is transmitted and the RX reads the channel frequency response $H(f)$ directly. Richer information (the full response), but OFDM is sensitive to carrier offset.
- **Diagnostic** (CW) — a single tone used to measure SNR, the carrier-frequency offset between the two radios, and data-path health. It does not characterize multipath.

How each mode works

CW — continuous-wave tone (diagnostic)

TX: loops a single clean 1000 Hz tone (offset from the NCO), 40 ms period with an integer number of cycles so the loop is perfectly phase-continuous.

RX: FFT the capture; find the tone (parabolic-interpolated peak) and measure power, in-band SNR, and the **carrier-frequency offset** versus the expected 1000 Hz — this is where the -3.15 Hz between-radio offset was read. It also scans for phase discontinuities and amplitude dropouts to confirm the data path is glitch-free.

Use it for: is the link and the plumbing healthy, what is the SNR, and what is the offset that OFDM has to fight.

LFM chirp — linear-FM sweep (matched filter)

TX: a 24 kHz-wide frequency sweep lasting 20 ms, then immediately the next, back to back. In a 5 s grab the RX sees about **250 sweeps** — not a single burst.

RX: coarse-sync to the first sweep, then **pulse-compress (matched-filter) every one of the ~250 sweeps** against the known chirp. Each compression collapses a 20 ms sweep into a sharp spike = one channel impulse-response snapshot. Compression gain = time-bandwidth product = $24000 \times 0.020 = 480 \approx +26.8 \text{ dB}$ (the OTA number that lifted -7 dB raw to $+19.8 \text{ dB}$).

Result: stack the ~250 snapshots into a power-delay profile (delay spread); watch each tap wobble across snapshots for Doppler. Carrier-offset tolerant.

DSSS/PN — direct-sequence spread spectrum (matched filter)

TX: loops a 255-chip BPSK m-sequence at 24 kchip/s (about 10.6 ms per code period), continuously.

RX: matched-filter each code period against the known sequence to get the impulse response. Processing gain = code length = $255 \approx +24 \text{ dB}$ (the OTA number). Stack over time for delay spread and Doppler.

Use it for: the same channel metrics as the chirp via a different probe; this is the waveform the OTA report recommends building the real data link on.

Comb — multitone (frequency-domain probe)

TX: loops a 40 ms block of 241 equally-spaced tones across 6 kHz, Schroeder-phased to keep the peak-to-average ratio low.

RX: FFT each 40 ms block and read the complex amplitude at each of the 241 tone bins — that is $H(f)$, the channel frequency response, measured directly (the known tone phases are removed first).

Result: coherence bandwidth two ways: spaced-frequency correlation of $H(f)$, and $\text{IFFT}(H)$ for the delay spread.

OFDM — pilot-loaded symbols (frequency-domain probe)

TX: loops one OFDM symbol — 512 QPSK-loaded subcarriers, 4096-point FFT, with a 5.3 ms cyclic prefix. Each symbol is 128 ms, so a 5 s grab is about **39 symbols**.

RX: cyclic-prefix sync to the symbol boundary, then **measure and remove the carrier-frequency offset** (blind, fresh from each capture), then per symbol FFT and divide by the known pilots to get $H[k]$. Yields the full frequency response plus a QPSK constellation / EVM data-fidelity readout.

Note: OFDM is the offset-sensitive mode — an uncorrected few-Hz offset collapses the constellation, which is why it failed the first OTA test. The offset is now measured per capture (it differs every test and drifts all day) and reported as “OFDM CFO removed: +/-X Hz”.

Quick reference

Mode	TX loops	RX recovers via	Gives you	Offset-tolerant?
CW	1 kHz tone	FFT peak	SNR, carrier offset, data-path health	n/a (it measures it)
LFM chirp	20 ms sweeps, back to back	matched filter x~250	impulse response, +26.8 dB gain	Yes
DSSS/PN	10.6 ms PN code	matched filter	impulse response, +24 dB gain	Yes

Mode	TX loops	RX recovers via	Gives you	Offset-tolerant?
Comb	40 ms, 241 tones	read $H(f)$ at bins	frequency response, coherence BW	Yes
OFDM	128 ms symbol	FFT / known pilots	freq response + constellation/EVM	Now corrected

Generated for the HL2 HF Sounder (main_sounder.py / channel.py / waveforms.py). Captured data and analysis files are saved to the capture\ folder. Self-test note: Duplex loopback shares one clock, so it cannot validate OFDM carrier-offset behavior — only an over-the-air test with the far station can.